The Spectre Knot

One tile, front and back

Front: the knot tile. Back: what it is. Edge length 41 mm.



The Spectre Knot

A single-strand Celtic knot on an aperiodic monotile

The tile is the straight-edged form of the spectre, discovered in 2023 by David Smith, Joseph Samuel Myers, Craig S. Kaplan and Chaim Goodman-Strauss: a single shape whose copies cover the whole plane, but only in patterns that never repeat.

The knot, the Spectre Knot, is the ribbon on the front. Justin Roughley came up with the original idea of looking for a mono-stranded aperiodic knot, and then used AI to search for it (2026). It is the same on every tile, yet across the nest tiling, grown outwards from one tile, its fourteen pieces join into **one endless strand:** a single Celtic knot through every tile, with no closed loops and no second strand, weaving over and under all the way, and never repeating.

No repeating tiling can do this: its symmetries would have to slide the one strand along itself in two independent directions at once, which a single line cannot do. It takes a tiling that never repeats, and the design is about as simple as it can be, with just one crossing inside the tile.

Why it works: the tiles group into supertiles of 9, 71, 559, 4,401 tiles and on for ever, and each larger supertile gathers the ribbon pieces onto fewer strands. A computation up to 133 million tiles, with the spectre paper's theorem that every level is built by the same rule, shows that every piece ends up on the same one. In every spectre tiling the ribbon never closes into a loop (the Endless Ribbon Theorem); one strand (the Ariadne Theorem) is proved for the nest tiling only, and computer checks suggest some others have three. The proof was made with AI and has not been checked by a mathematician.

Justin Roughley · 7 Oct 2026

1. Cut round the outer black line, as one piece.
2. Fold along the dashed line, printed sides out.
3. Glue the halves back to back.

Make several and lay them edge to edge: the ribbons meet where the edges touch and cross there at right angles.